



ENT-164 · Intro to Making

Tufts
UNIVERSITY

Derby Entrepreneurship
Center at Tufts

CLASS 2 · WEEK 2 · FALL 2026

Hand Making and Robot Challenge Kickoff

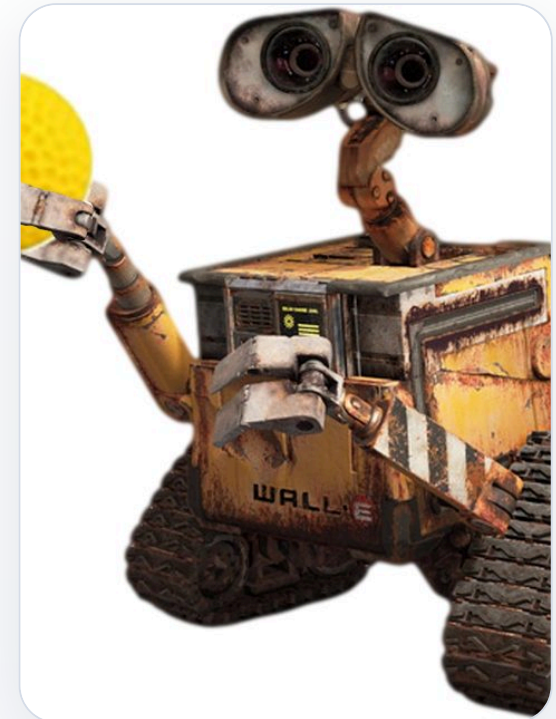
Today we kick off the Robot Challenge: the rules, your team, the electronics in your kit, and the hand-making skills, foam core and clay, you'll use to prototype your robot.

Thu, Sep 17

JCC 301 / Nolop

Robot Challenge kickoff

Teams announced today



Move as many balls as you can in 5 minutes

Your robots must be designed, built, and programmed by your team.

TO SUCCEED

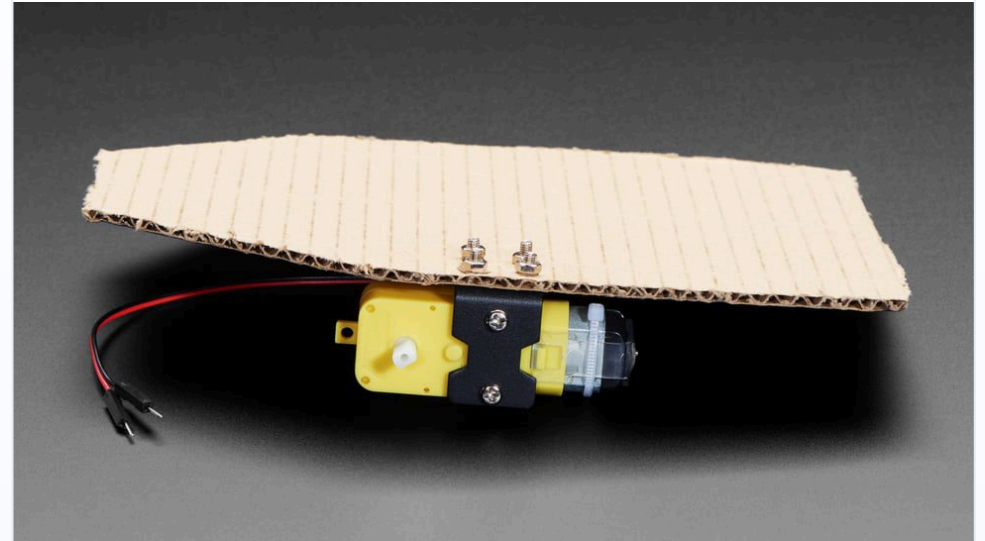
You'll combine creativity, prototyping, mechanical design, and coding. The whole maker toolkit in one challenge.

Creativity

Prototyping

Mechanical design

Coding



How the race **works**

Read these twice. Every design decision follows from them.

SCORING

Every ball counts

- 1 point per ball moved across the line
- Maximum 10 balls per trip
- Balls are dropped off in a standard container

THE COURSE

5 minutes, 10 feet

- Move as many balls as possible in 5 minutes
- Loading and drop-off zones are 10 feet (~3 meters) apart

YOUR ROBOT

Hands off during the race

- Designed, built, and programmed by your team
- You cannot touch the balls or robot during the race
- Robot and loading mechanism must fit in your team's project box

This is about more than **winning**

01

Experiment quickly

Try ideas fast and learn from what breaks.

02

Build foundations

Practice sketching, prototyping, electronics, and coding.

03

Collaborate

Learn to work as a team on a shared build.

04

Solve openly

Find creative answers to open-ended problems.

THE PAYOFF

At the end, you'll not only have a robot to show off. You'll have the confidence and maker skills to take on your **final team project**.

Upcoming workshops & key dates

You don't have to wait for workshops to apply these techniques. Learn from others, research online, or ask Nolop staff.

TODAY

Ideation and Prototyping

Sketch · Foam core

SEP 24

From Physical to Digital: 2D

2D design · Laser cutting

OCT 1

From Physical to Digital: 3D

Digital sculpting · 3D design · Slicing

OCT 8

Basic Electronics and Microcontrollers

Micro controller · MicroPython · Motors

OCT 15

AI Coding Agents

Networking · Remote control

OCT 22

Final Robot Assembly and Testing

Finalize · Test

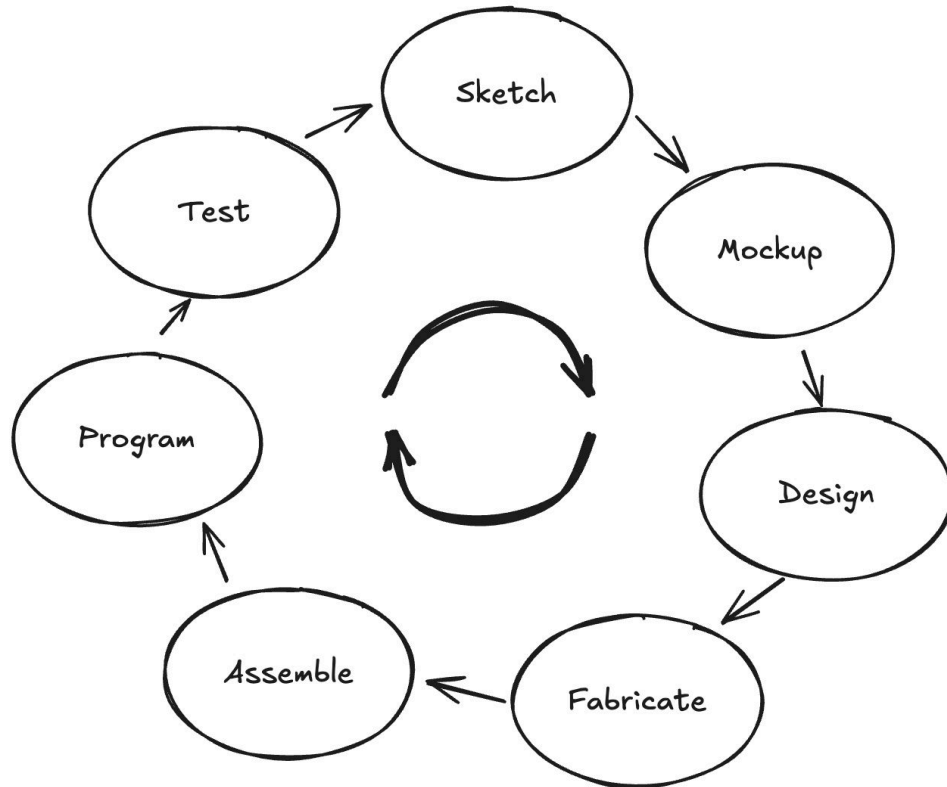
OCT 29

Robo Challenge and Final Project Kickoff

Race day. Put your robot to the test.

Product development is **not linear**

Expect to loop. Testing sends you back to the sketch, again and again.



The loop

Sketch → mockup → design → fabricate → assemble → program → test. Then test again.

Going backwards is progress

When a test fails, you go back a step, or all the way back to the sketch. That's how prototypes get better.

Motors and wheels **get you moving**

DRIVE

DC gearbox motor • 2x

- Your powered wheels. One per side
- 3mm mounting holes for screws



DC gearbox motor

SUPPORT

Caster wheels

- Free-rolling wheels that carry the weight
- Let the robot pivot and turn smoothly



Caster wheel



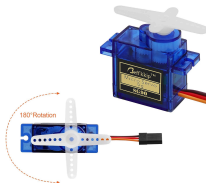
Motor + wheel



Motors + casters in the wild

Servo motors for precise moves

Servos rotate to a specific angle. Use them for steering, grippers, and your ball-dispensing mechanism.



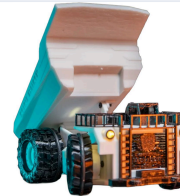
SG90 9g micro servo

The servo in your kit.



Servo horns

The arms that come with it. Pick one for your mechanism.



Wired up

Servos connect to the breadboard.



In the wild

Legged robots run on servos.

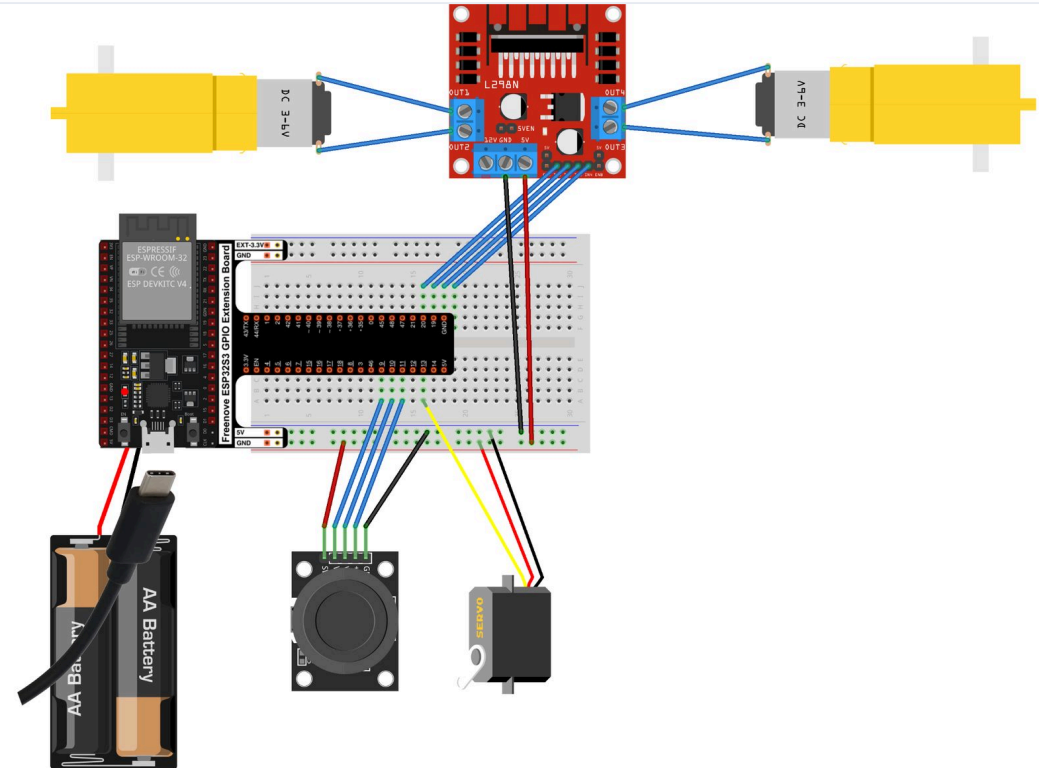
The rest of the electronics

Everything on this diagram comes in your kit.

WHAT EACH PART DOES

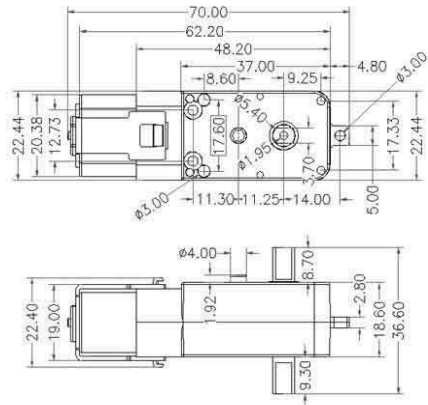
The control chain

- Motor driver. Controls speed and direction of both motors
- Breadboard. Connects the wires to the microcontroller
- Joystick with push button, your remote control
- Battery pack, powers the microcontroller
- Microcontroller. The brain; keep its USB-C port accessible to upload code



Component details & wiring

During early design, don't worry about exact connections. Plan the space and the wire routes.



EXACT PARTS

Search the data sheet

- DC gearbox motor. "TT motor"
- L298N dual H-bridge motor driver
- KY-023 control stick
- SG90 9g micro servo
- Freenove ESP32 WROOM
- TCRT5000 infrared sensor (optional)

PLAN YOUR WIRING

Leave room for these runs

- Both motors → motor driver
- Motor driver → breadboard
- Servos → breadboard
- Joystick (long wires) → breadboard
- Battery pack → microcontroller
- USB-C port accessible for code uploads

TIP: Try to hide wires for a better finish.

Get inspired, then research solutions

Study existing mechanisms

How do other makers move balls, dispense objects, and steer small robots?

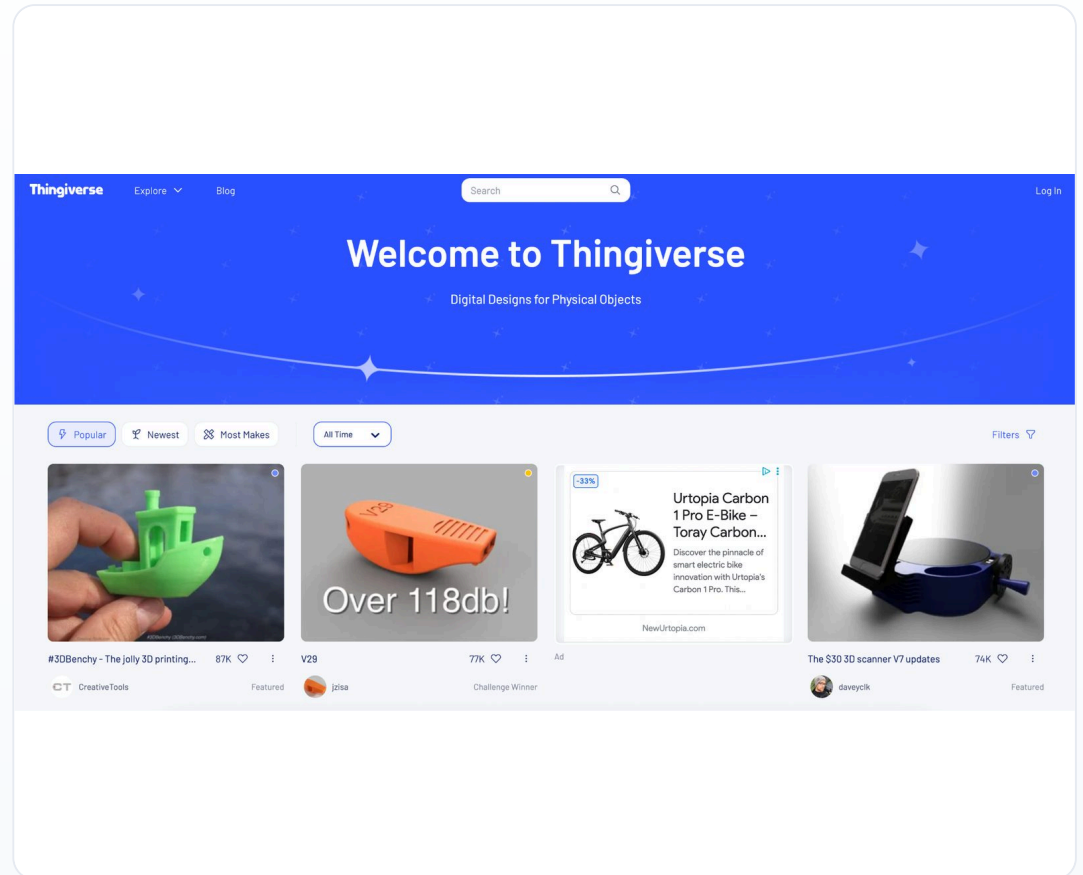
Borrow ideas, not parts

Adapt what you find to the components already in your kit.

Thingiverse

Build videos

Nolop staff



From brainstorm to prototype

Steps you'll take today and in the next workshop.

FIRST, ON YOUR OWN

Review and sketch

- Review the rules and constraints
- Review the parts in your kit
- Sketch ideas individually. Aim for several concepts

THEN, AS A TEAM

Converge and build

- Brainstorm the robot and a ball-dispensing mechanism
- Converge on one or more concepts to prototype
- Assign tasks and build with foam core, clay, glue, and screws
- Present your prototype(s) next class, Sept 24

ENT-164 project teams

Orange

Robot Challenge team

Blue

Robot Challenge team

Red

Robot Challenge team

Green

Robot Challenge team

TEAM ASSIGNMENT

Due Sept 24. Present your initial prototype to the class (5 minutes per team at Nolop).

Work safely, every time

01

Ask first

Follow instructor directions and ask before using unfamiliar tools.

02

No food or drinks

Keep them away from tools, electronics, and materials.

03

Tidy space

Keep your workspace organized and clutter-free.

04

Closed-toe shoes

Wear them at all times in the lab.

Hand tools & prototyping materials

CUTTING

Blades and mats

- Cut away from your body and keep hands clear of blades
- Use cutting mats for X-Acto and utility knives
- Foam core, clay, and cardboard are safe to prototype with, but still use blades carefully

IN THE LAB

Dress and behave for the shop

- Return all tools to their designated location after use
- Wear safety glasses when cutting, soldering, or using power tools
- Gloves only for hot materials or chemicals. Never around rotating tools
- Tie back long hair and avoid loose clothing or jewelry near machines

POLICY Review Nolop's safety policy here: nolop.org/safety

What is foam core?

Light and rigid

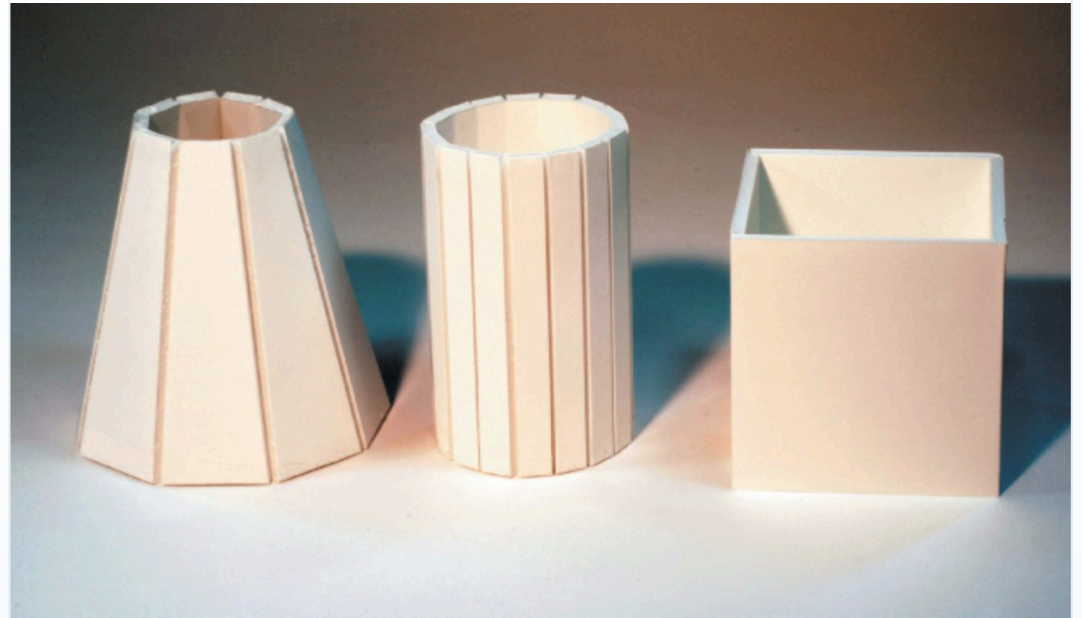
A Styrofoam core laminated between two layers of smooth paper.

Better than cardboard

Easier to work with, and it produces neater, stronger models than corrugated cardboard.

Shape almost anything

Most geometric shapes can easily be created in foam core.



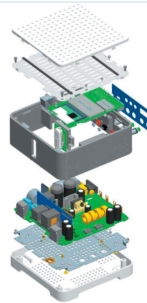
Component placeholders **get feedback**

Before the electronics go in, mock up the body with simple stand-ins for the real parts.



Component placeholders

Blocks stand in for the real parts so you can test fit.



Exploded view

Plan how the layers and parts stack together.



Teardown study

See how a real product is assembled inside.

Cutting: sharp blade, three passes

HOW TO CUT

Mat, ruler, fresh blade

- Always cut on a cutting mat with a sharp X-Acto knife
- Replace blades at the slightest hint of wear
- Straight cuts are done with a metal ruler
- Curves are cut carefully by hand



Pass 1

Top paper layer



Pass 2

Through the foam

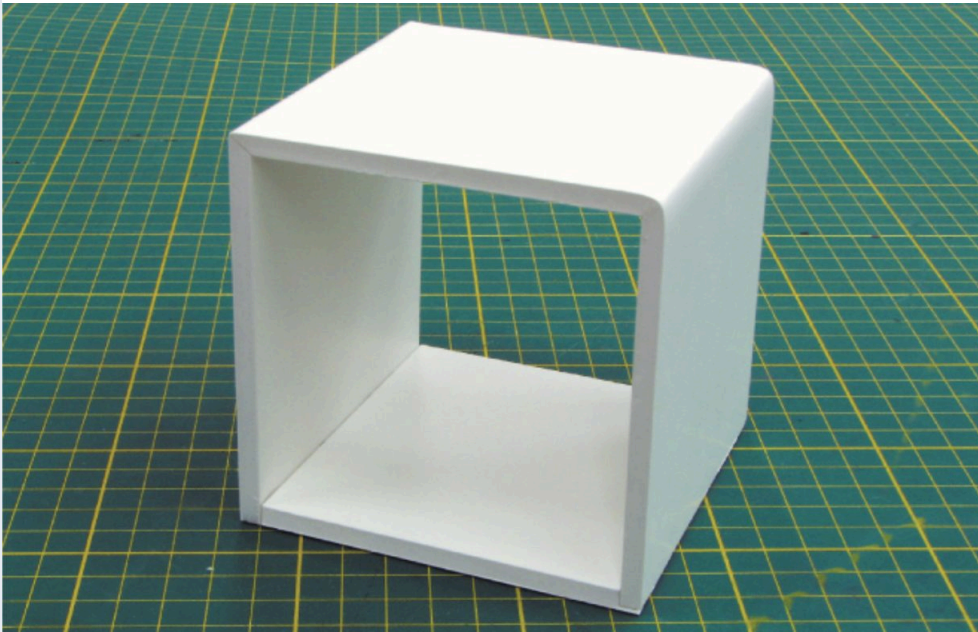


Pass 3

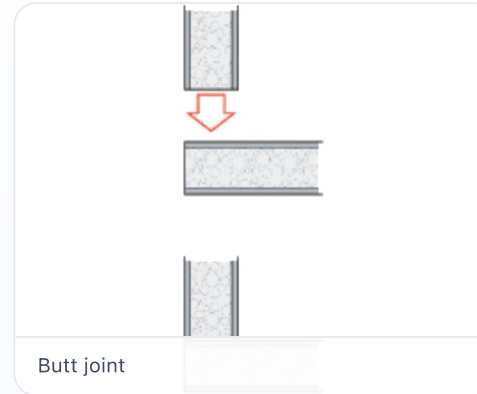
Bottom paper layer

Corners: four joints

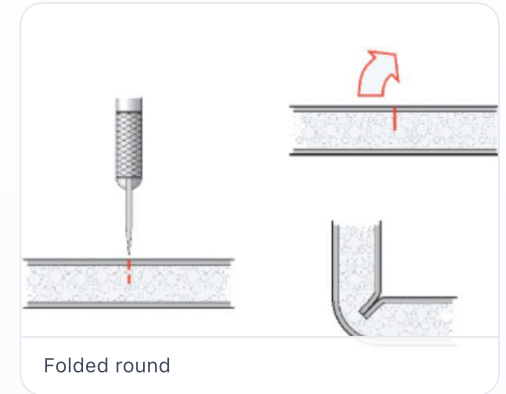
The rabbet joint hides the edge of the board and makes the strongest, cleanest corner.



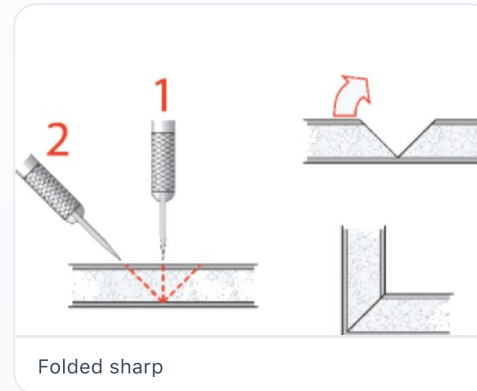
A finished foam core corner



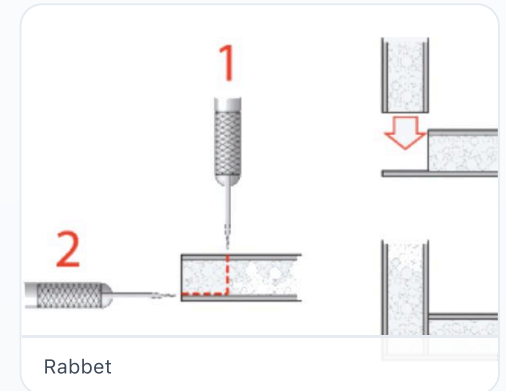
Butt joint



Folded round



Folded sharp

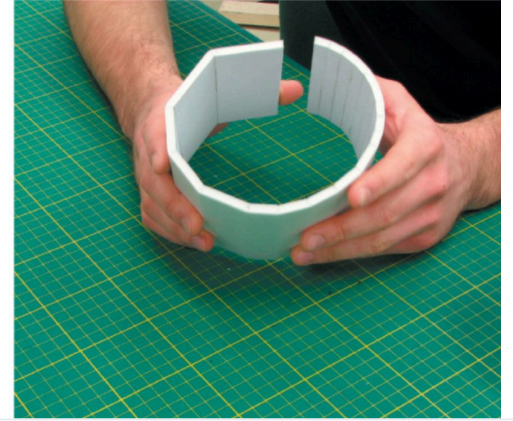
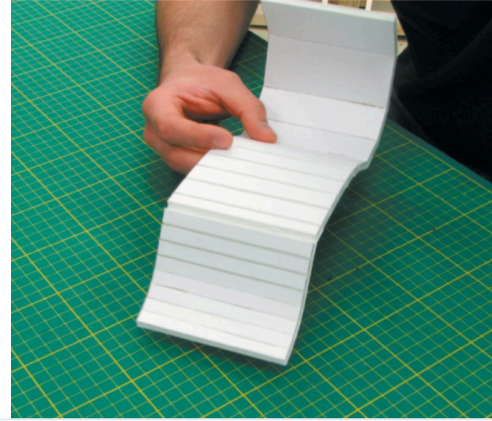


Rabbet

Curves: cut the outside, notch the inside

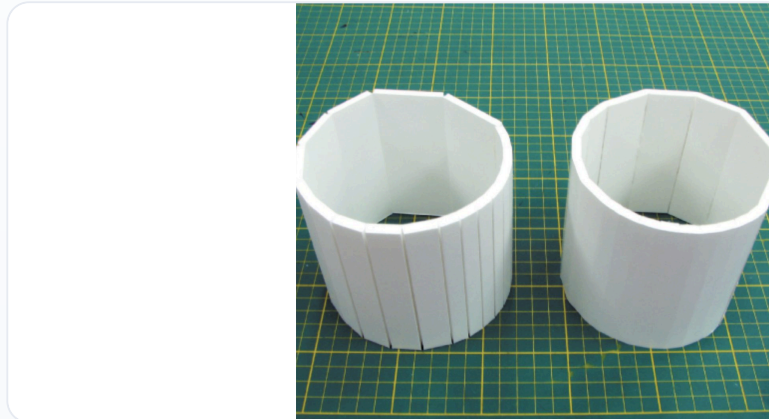
Outside

Make cuts on the outside of the curve to release the paper layers.



Inside

Cut V-notches on the inside so the board bends smoothly around the curve.



Train ticket kiosk in foam core

A full-size model built with the same techniques you saw. From a rough sketch to a finished kiosk.

What it takes

Paper templates, precise cuts, rabbet joints, curved surfaces, and a clean finish.

Techniques used

Sketch

Templates

Rabbet joints

V-notches

Hot glue

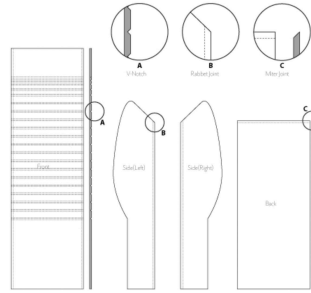
Decals



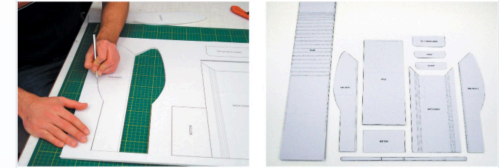
Steps 1–6: sketch to first assembly



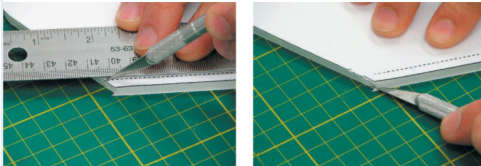
1 • Sketch



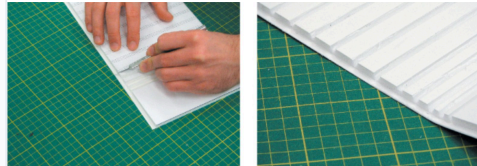
2 • Draw template patterns



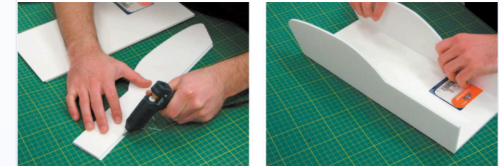
3 • Glue template, cut



4 • Trim the rabbit

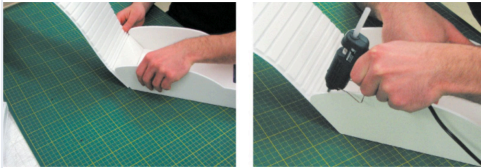


5 • Rabbit + V-slot



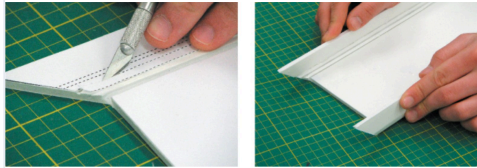
6 • Hot glue

Steps 7–12: curves, supports, finish



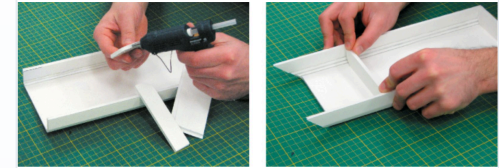
7 • Attach the curved surface

Hot glue along the seam.



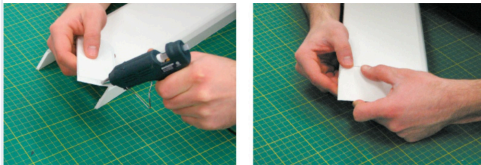
8 • Curved corners

Three V-notches per corner.



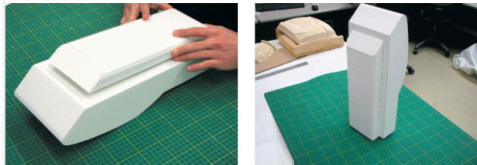
9 • Internal supports

Add strength where it counts.



10 • Curved top

Rabbit joint along the roof.



11 • Attach the back

Hot glue the back panel.



12 • Decals + panel

Finish the front with graphics.

Clay: organic forms

A naturally expressive, sculptural material, and the first step toward a 3D-printed part.

WHY USE CLAY?

Sculpt, then scan

- A naturally expressive and sculptural medium
- Easily manipulated by hand and simple tools
- The model-making material of choice for complex, organic forms
- Clay models can be scanned and refined in software for 3D printing



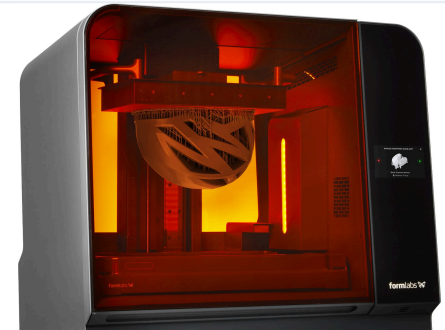
Modeling clay



Example product



Sculpted model



Refined and printed

Five steps from sketch to scan



1 • Sketch model

Draw the form first.



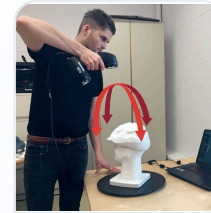
2 • Apply clay

Build up clay on the fixture.



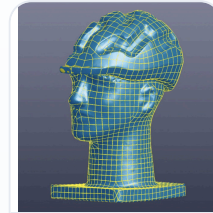
3 • Sculpt

Shape and smooth the model.



4 • Scan

Digitize the model.

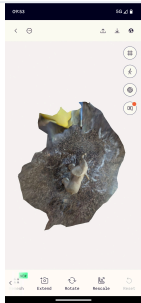


5 • Post-process

Clean up the digital model.

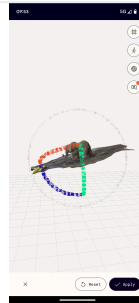
Install Polycam, **scan your model**

Use your phone's camera to scan and digitize your clay model.



Scan

Walk around your model and capture it.



Process

Let the app build the 3D model.



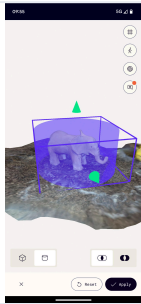
Rotate

Turn the model to prepare it for export.

THE WORKFLOW Install the **Polycam** app → scan your model → rotate and crop it → export. The next slide converts it to STL.

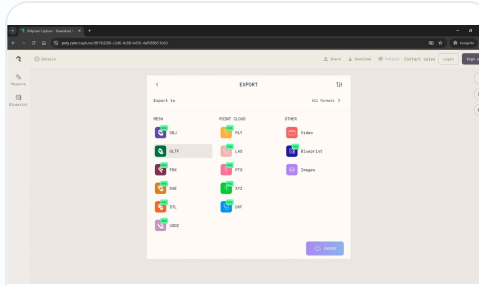
Crop, export, convert to STL

Clean up the scan, then convert it so it can be 3D printed.



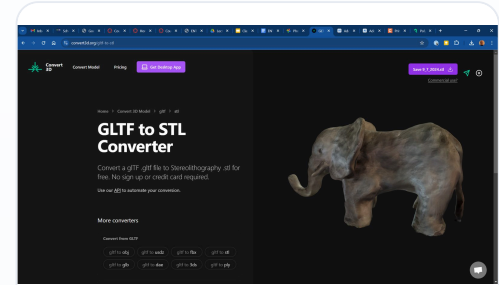
Crop

Remove the desk and background around the model.



Export

Save the model as a file you can convert.



Convert

Turn it into an STL for 3D printing.

CONVERT

Use convert3d.org/gltf-to-stl to convert your exported model to STL.

BEFORE YOU GO

Questions?

Then it's time to sketch. Next class: present your initial prototype at Nolop and start turning your sketches into 2D designs.

Team assignment: initial prototype presentation. Due Sept 24, 5 minutes per team at Nolop.

WHERE TO FIND US

Office hours

go.tufts.edu/meet_tvdv

Email

thomas.van_de_velde@tufts.edu

Course materials

tuftsmaker.github.io/ENT-164